POLICY BRIEF · GREEN ALIBI

🌿 Can Satellite Fluorescence Data Speed Up Drought Relief for Marathwada's Farmers?

Testing whether Solar-Induced Fluorescence (SIF) detects agricultural drought stress earlier than the indicators currently used for official drought declaration and PMFBY crop-insurance payouts.

Sakshi D. Maske — Independent Geospatial Researcher · Marathwada, Maharashtra

The Problem

India's drought declaration and PMFBY crop-insurance payout process relies mainly on rainfall records and NDVI — a satellite index measuring how "green" a crop canopy looks. Both indicators register stress only after visible physical damage has occurred, by which point the season's window for meaningful relief is closing. In Marathwada, this delay has repeatedly meant relief and payouts arriving after farmers most needed them.

What Was Tested

SIF is grounded in a plant's internal photosynthetic process rather than its outward appearance, so it should register stress earlier than NDVI. This study tested that directly, using GOSIF v2 and cloud-screened MODIS NDVI over Marathwada's eight districts across two drought years (2015, 2018) and one normal year (2020), independently cross-validated against CHIRPS rainfall.

Study at a Glance

3/3 years SIF led NDVI	**100%** bootstrap replicates confirm lead	**r = 0.837** SIF–rainfall correlation
~7 weeks satellite lead vs. official 2018 declaration	**3–4 days** SIF's edge over NDVI specifically	**6/6** Moran's I checks confirm real spatial pattern

What Was Found

SIF leads NDVI, consistently. Across all three years studied, SIF's seasonal decline began before NDVI's — confirmed by two independent statistical methods, and shown by bootstrap resampling (2,000 replicates/year) to hold in every resampled scenario tested.

The bigger opportunity is elsewhere. Comparing 2018's satellite signals against the actual date Maharashtra officially declared drought (31 October 2018) shows both SIF *and* NDVI crossed their stress thresholds roughly **seven weeks** earlier. SIF's edge over NDVI itself (3–4 days) is real but small next to this much larger gap between any satellite signal and the current institutional process.

Drought severity does not simply make the lag bigger. A natural hypothesis — that more severe drought produces a larger SIF–NDVI gap — was tested directly and was **not supported**. This negative result is reported directly, not adjusted or hidden.

The stress pattern lines up with rainfall independently, and this spatial correspondence is confirmed statistically real (Moran's I, all six year × variable checks significant) rather than a byproduct of neighboring districts sharing weather.

Policy implication. PMFBY assessment already incorporates satellite imagery for loss estimation — the infrastructure for a satellite-triggered early-warning step already exists in principle. The evidence here points to a two-part opportunity, in order of scale: **(1)** shortening the roughly seven-week gap between when satellite data of any kind already shows stress and when the official declaration process catches up, and **(2)** layering SIF's smaller, physiologically-earlier refinement on top of that larger improvement.

What This Study Does Not Yet Establish

- This is a three-year proof-of-concept, not an operational system, and covers one region.
- It has not been validated against ground-level crop yield or crop-loss data.
- GOSIF (the SIF product used) is partly modeled from MODIS data, so it is not fully independent of the NDVI it is compared against.

Full methodology, statistics, and complete limitations: Research_Paper.md. Full development history, including every correction made along the way: Development_Log.md. Interactive dashboard and all figures available on request.